

Where Learning Helps and Where It Doesn't (v10)

Stage attribution · learned infeasibility test · scaling wall at 60×150

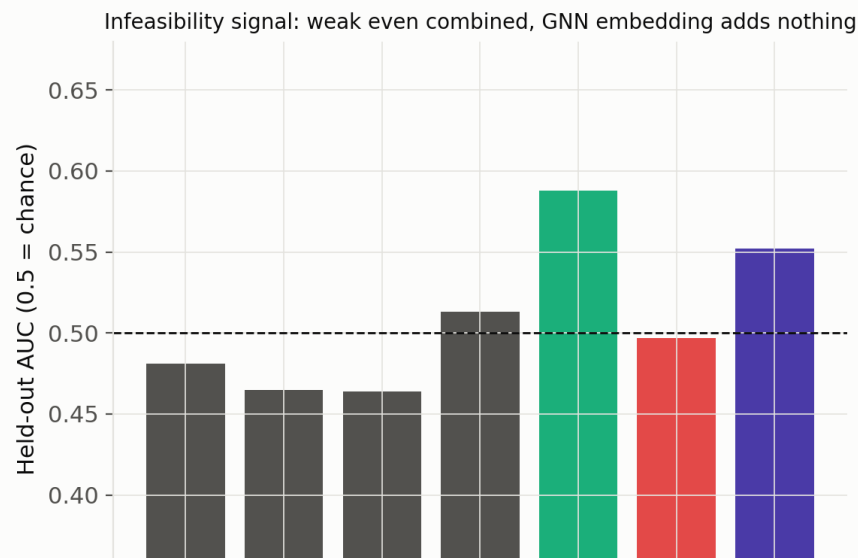
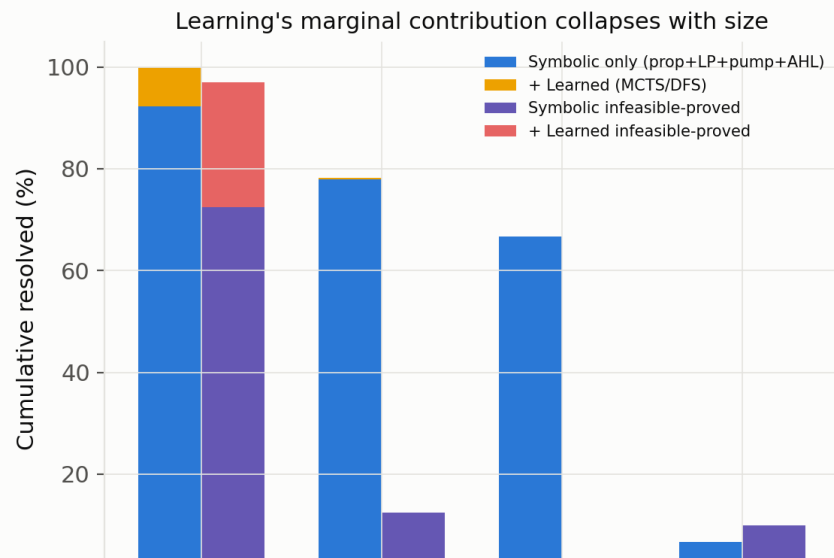
neuroMCTS project · July 2026

Learning's marginal value converges to exactly zero with size

10×25: learning adds +7.6-7.75pp solved, +24.5pp infeasibility proofs — meaningful

20×50: shrinks to +0.4pp solved, +0pp proofs

40×100 & 60×150: learning contributes literally ZERO instances



Sixth negative: infeasibility signal is weak, GNN embedding is null

New feature: AGGREGATE stats from 15 pump restarts + 8 cheap AHL tries per instance

Combined scalar features: AUC 0.588 (single features: 0.46-0.54) — real but weak

Pretrained GNN embedding alone: AUC 0.497 — exactly uninformative

At best precision: 1 in 3 flagged infeasible is actually feasible — unusable at FN=0

Discovered scaling wall: an accuracy cliff at 60×150

N=40 statistical spot check at 60×150 (never trained/evaluated before)

Solution accuracy: 66.67% (40×100) → 6.67% (60×150) — a cliff, not a gradual decline

Mean time: 22.0s → 108.0s; learning contributes zero here too

Small-N as instructed (not exhaustive) — sufficient to LOCATE the boundary

Conclusion: three precise, falsifiable statements

1. Learning helps only at the smallest size, inside an already-narrow residue
2. Learning cannot classify feasibility — not even from its own GNN representation
3. The symbolic AHL attack that does the real work has its own wall near $n \approx 150$

State of the art: strong to $n \approx 100$, a measured wall beyond it